SYSTEMS VULNERABILITY ASSESSMENT

CYS330

OPERATING SYSTEM

SCHEDULING ALGORITHMS

HARDWARE TROJANS (HT)

- A **hardware trojan** is a malicious modification of an integrated circuit's circuitry, designed to disrupt operations, steal data, or cause physical harm or death.
- These modifications can be inserted during the design or manufacturing stages, often through untrust third-party components or intellectual property cores.

CHARACTERISTICS OF HARDWARE TROJANS

1. **Size** – This refers to the number of components added or modified. Trojans can range from **small transistors** to **large circuits**.
2. **Distribution**

i. **Tight Distribution** – Trojan components are located close together in the layout.

ii. **Loose Distribution** – Components are dispersed across the chip making detection harder

CHARACTERISTICS OF HARDWARE TROJANS (CONTINUES…)

3. **Structure** – Changes in the physical layout can alter the chips (IC) dimensions or component placement.
4. **Type**

i. **Functional Trojans –** These involve adding or deleting components such as transistors or logic gates to modify chip's functionality.

ii. **Parametric Trojans** – These involve modifying existing components such as thinning wires or weakening transistors, to affect the chip's reliability or performance.

IMPACT OF HARDWARE TROJANS

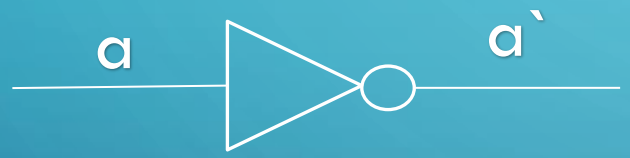
1. Data Breaches and Intellectual Property Theft
2. System Malfunction
3. Performance Degradation
4. Reputational Damage and Financial Costs
5. Denial – of – service (DoS) Attacks

MITIGATION AND REMEDIATION TECHNIQUES OF HT

1. Detection Techniques
2. Collaborative Methods
3. Message Authentication Codes (MACs)

LOGIC GATES

1. NOT



TRUTH TABLE

INPUT (a)	OUTPUT (a`)
1	0
0	1

LOGIC GATES (CONTINUES…)

2. AND



TRUTH TABLE

INPUT (a)	INPUT (b)	OUTPUT (ab)
0	0	0
0	1	0
1	0	0
1	1	1

LOGIC GATES (CONTINUES…)

3. OR



TRUTH TABLE

INPUT (a)	INPUT (b)	OUTPUT (a + b)
0	0	0
0	1	1
1	0	1
1	1	1